
SaaS Lead Generation & Revenue Intelligence Pipeline

BILL SWIFT

An end-to-end automation project for Bill Swift, a usage-based billing SaaS company

A portfolio project by Praveen Sambasivam

What is Bill Swift?

Bill Swift is a fictional SaaS company built as the business scenario for this portfolio project — modeled on Chargebee, a real subscription billing and revenue management platform used by SaaS companies. Bill Swift gives every part of this pipeline a reason to exist: a webinar, a scored lead, a CRM record, and a revenue forecast that a real billing-SaaS company's sales and finance teams would actually use.

What is this project about?

This project is an automated, end-to-end SaaS lead-generation and revenue-intelligence pipeline. From the moment someone registers for a webinar to the moment their deal closes, every step — scoring, CRM sync, email nurture, revenue prediction, and reporting — runs without manual intervention.

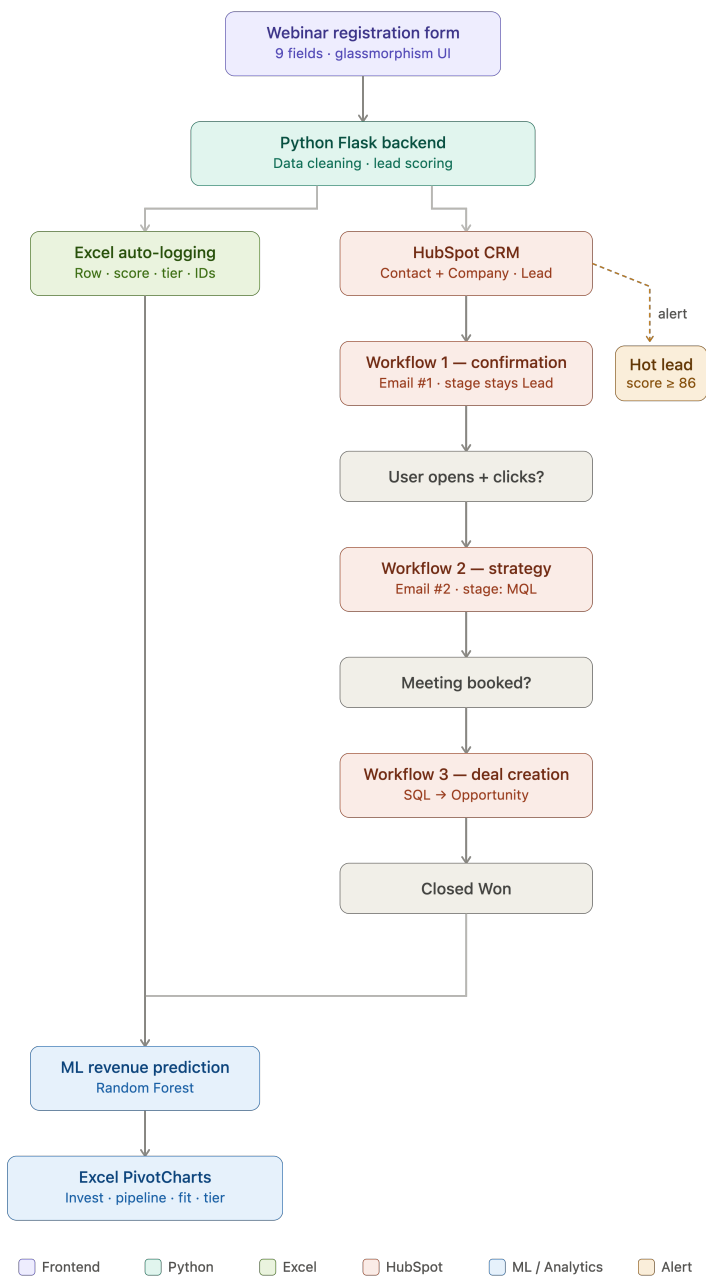
What problem does this project solve?

Every SaaS company faces the same gap between a new lead and a closed deal — leads come in faster than anyone can manually qualify, track, or follow up with. For Bill Swift, I built a system to close that entire gap automatically:

- **Lead qualification** — a 6-signal, 100-point scoring engine scores every lead automatically based on the details captured at signup (company size, transaction volume, current billing setup, and more), so sales isn't guessing who to prioritize.
- **CRM and data consistency** — every lead syncs to Excel and HubSpot simultaneously with matching IDs, keeping one reliable source of truth instead of two drifting records.
- **Hot lead response time** — an automated alert flags any lead scoring above 85, so high-value prospects aren't missed while sitting in a queue.
- **Structured follow-up** — tiered nurture workflows move leads through Lead → MQL → SQL → Opportunity → Closed Won without manual handoffs.
- **Revenue visibility** — an ML revenue-prediction model feeds CXO-ready PivotCharts, showing leadership where the pipeline is strongest and where to invest.

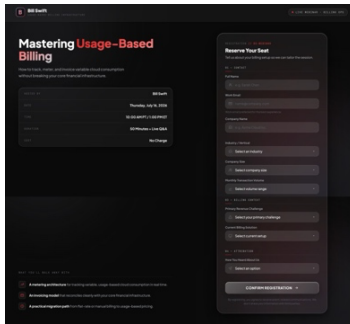
Pipeline Overview

Here's how a lead moves through the system, end to end:

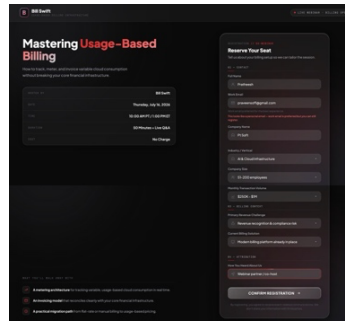


1. Webinar registration form

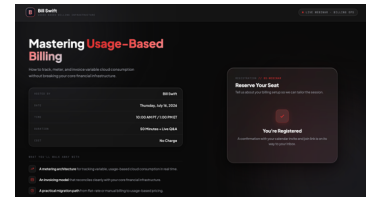
Lead-gen entry point — a webinar signup form that captures contact, company, and billing context in one step.



Full webpage — event details and registration form



Form mid-fill — personal email soft warning shown



Success screen — confirms the pipeline is live

2. Python Flask backend

Receives the form data, cleans it, and triggers scoring — all in one request.

3. Lead scoring engine

Scores each lead out of 100 across 6 signals; sorts into Hot / Warm / Cold.

Signal	Max
Monthly Transaction Volume	28
Primary Revenue Challenge	24
Current Billing Solution	19
Company Size	14
How You Heard About Us	10
Email Type	5

Tiers: 0-35 Cold · 36-65 Warm · 66-100 Hot

Six weighted scoring signals feed a 100-point scale, split into three tiers.

4. Excel auto-logging

Auto-appends every lead to Excel with score, tier, and IDs — no manual entry.

Timestamp	Name	Email	Company	Company	Industry	Industry	Company	Monthly	Primary	Current	Heard From	Lead Score	Lead Tier	HubSpot	HubSpot	Predict
2026-07-01	Praveen Sai	praveen.dur	Prvn Tech	Personal Er	edtech		201-1000	250k-1m	manual-in	stripe-payp	google	58	Warm	51244505	53152383	152837.3
2026-07-01	Sarah Chen	sarah@veri	Vertex AI	vertexai.io	ecommerce		201-1000	50k-250k	usage-mete	in-house	newsletter	65	Warm	51244541	53157054	227258
2026-07-01	David Ross	dross@saa	SaaSflow	saasflow.cc	proptech-hot		201-1000	250k-1m	usage-mete	in-house	partner	75	Hot	51244342	53153833	271457.6
2026-07-01	Elena Rostic	elena@cyb	Cybercore	cybercore.ai	fintech		11-50	10m+	research	stripe-payp	google	60	Warm	51245502	53149321	184046.1
2026-07-01	Elena Rostic	elena@cyb	Cybercore	cybercore.ai	ai-cloud		11-50	250k-1m	research	legacy-ente	referral	45	Warm	51245502	53149321	126484.1
2026-07-01	Marcus Var	marcus@p	Paybolt	paybolt.fin	ecommerce		51-200	1m-10m	research	legacy-ente	newsletter	51	Warm	51238017	53154755	157615.1
2026-07-01	Chloe Tana	c.tanaka@	Edulearn	edulearn.ci	edtech		1-10	under-50k	rev-recogni	in-house	partner	47	Warm	51235872	53147174	180323.1
2026-07-01	Liam O'Con	liam@stres	Streamvib	streamvibe	digital-media		1-10	250k-1m	usage-mete	none-man	other	66	Hot	51242147	53154585	238010.9
2026-07-01	Amara Okai	a.okafor@	Nexthome	nexthome.	proptech-hot		51-200	10m+	research	stripe-payp	linkedin	65	Warm	51235194	53153292	193846.6
2026-07-01	Jacob Miller	j.miller@bi	Biotrack	biotrack.or	ai-cloud		1000+	50k-250k	manual-in	none-man	google	62	Warm	51244507	53149501	171892.2
2026-07-01	Pratish	praveensof	Pt Soft	Personal Er	ai-cloud		51-200	250k-1m	rev-recogni	modern-pli	partner	49	Warm	51238751	53152383	189583.2

Every submission appends a new row — score, tier, and HubSpot IDs included.

Creates Contact + Company in HubSpot; handles duplicate conflicts automatically.

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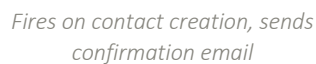
HubSpot Contacts — auto-created on every submission

HubSpot Companies — linked to matching contacts

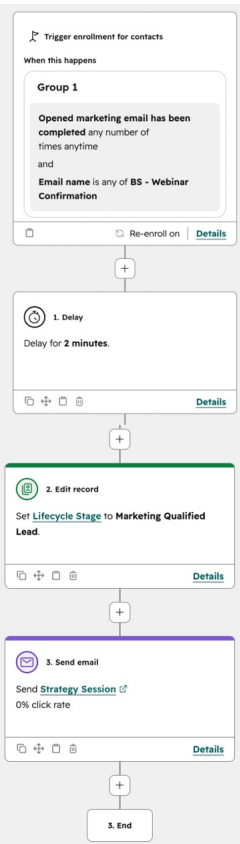
4 workflows: confirmation email → MQL upgrade → SQL/deal creation → hot lead alert.

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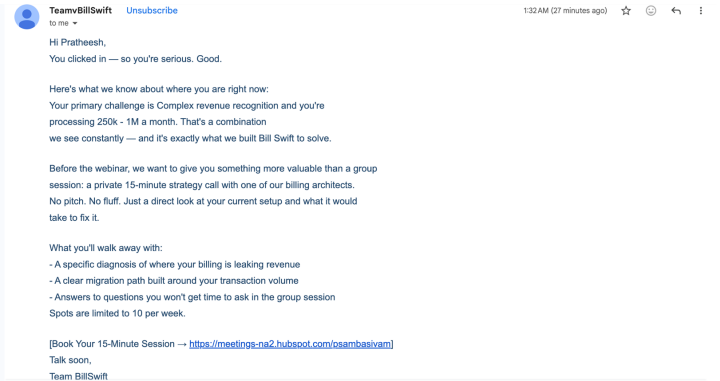
Workflow 1 — registration confirmation



Workflow 2 — MQL upgrade & strategy session

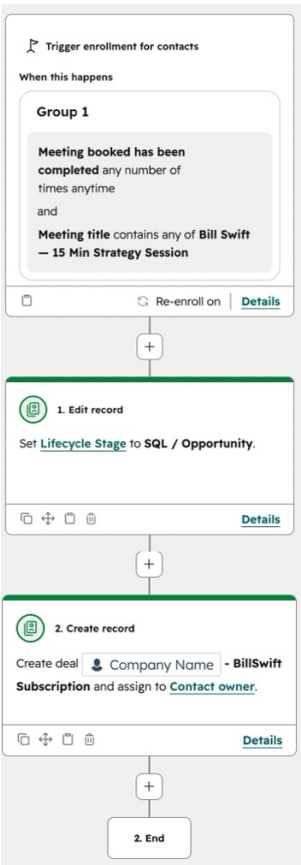


Email open triggers MQL upgrade,
sends Email #2



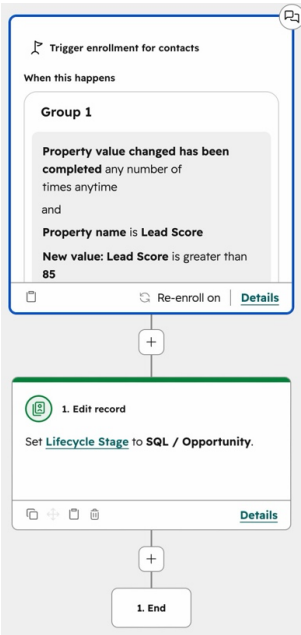
Email #2 — dynamic tokens populate real challenge + volume

Workflow 3 — meeting booked → SQL & deal creation



Meeting booked triggers SQL/Opportunity and auto-creates a deal.

Workflow 4 — hot lead alert (Lead Score > 85)



Lead Score above 85 triggers an instant SQL upgrade and sales alert — independent of the email sequence.

Meeting booked → deal auto-created in pipeline → progresses to Closed Won.



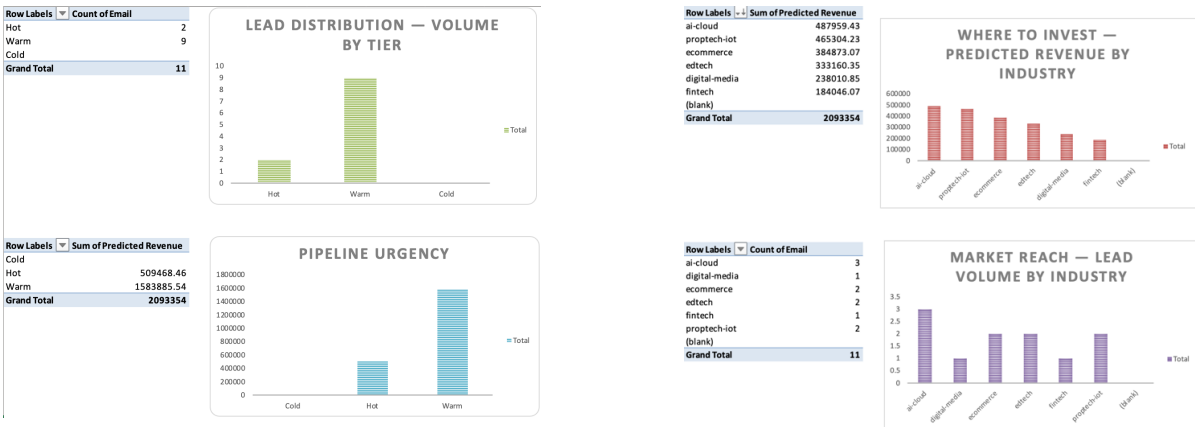
Random Forest model predicts contract value; written back to Excel.

200-row historical dataset used to train the Random Forest model at server startup.

Predicted revenue is written back into the same Excel row as the lead score.

9. Excel PivotCharts

4 live charts: revenue by industry, lead volume, tier distribution, pipeline urgency.



Lead Distribution by Tier (2 Hot · 9 Warm) and Pipeline Urgency

Where to Invest — revenue by industry, and Market Reach by volume

Project Summary

- Zero manual steps — from form submission to CRM contact, ML prediction, and email delivery, everything is automated.
- Server-side lead scoring — scores are calculated in Python and cannot be tampered with from the browser.
- Dynamic email personalization — Email #2 injects real CRM properties into each send.
- Full audit trail — every row contains a timestamp, cleaned data, lead score, tier, HubSpot IDs, and predicted revenue.